

Highlights

Short-Term Solar Irradiance Forecasting in Tropical Andean Climates: A Graph Neural Network Approach with Disentangled Uncertainty Quantification

Esteban Ladino Nieto

- GraphSAGE-LSTM for GHI forecasting, with an ablation isolating fixed vs. learned weighted k -NN spatial connectivity.
- Rigorous comparison against ResNet-LSTM, a flat-MLP baseline, and persistence at two climatically contrasting Colombian sites.
- A disentangled aleatoric/epistemic uncertainty-quantification layer via a Gaussian NLL variance head and SGLD posterior sampling.
- A reproducible GOES-16 benchmark for tropical Andean solar forecasting, a climate regime underrepresented in the literature.

Short-Term Solar Irradiance Forecasting in Tropical Andean Climates: A Graph Neural Network Approach with Disentangled Uncertainty Quantification

Esteban Ladino Nieto^{a,*}

^aDepartment of Industrial Engineering, Universidad de los Andes, Bogotá, Colombia

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ABSTRACT

Solar photovoltaic (PV) generation is expanding rapidly across tropical Latin America: Colombia alone has surpassed 1 GW of installed capacity and targets 9 GW by 2030 under its National Energy Plan (Unidad de Planeación Minero Energética (UPME), 2023). Reliable short-term forecasts of global horizontal irradiance (GHI) are a prerequisite for efficient operation of PV-integrated power systems, but point forecasts alone cannot support risk-aware reserve scheduling: operators additionally need to know *why* a forecast is uncertain — because the underlying cloud field is intrinsically unpredictable (aleatoric uncertainty), or because the model has seen too little similar data (epistemic uncertainty) — since the two sources call for different operational responses. In tropical countries such as Colombia, the forecasting problem is compounded by the high frequency and spatial heterogeneity of convective cloud systems, which can reduce surface irradiance by more than 80% within minutes and are poorly captured by coarse-resolution numerical weather prediction models. This paper proposes a graph neural network framework for short-term GHI forecasting from GOES-16 geostationary satellite imagery and couples it with a **disentangled uncertainty-quantification layer** that separately estimates aleatoric uncertainty, via a Gaussian negative-log-likelihood variance head, and epistemic uncertainty, via posterior weight samples obtained with Stochastic Gradient Langevin Dynamics (SGLD), to support robust, risk-stratified forecasting. We introduce **GraphSAGE-LSTM**, which treats each satellite patch as a graph of spatially connected pixels and performs neighbourhood aggregation via GraphSAGE before passing the resulting embedding sequence through an LSTM; we compare a fixed unweighted nearest-neighbour graph against a learned, distance-weighted variant with a tunable neighbourhood size to determine the most effective spatial representation. The model is evaluated against a ResNet-LSTM baseline, a flat-MLP spatial-ablation baseline, and persistence, at two Colombian sites representing contrasting climatic regimes: El Paso (César), a high-irradiance low-elevation tropical site, and Universidad de los Andes (Bogotá), a high-altitude equatorial mountain station with intense convective variability, across forecast horizons of 1, 3, and 6 hours. Preliminary single-seed comparisons indicate that graph-based spatial encoders are competitive with or superior to convolutional baselines at multi-hour horizons, while absolute skill remains markedly lower at the high-altitude site, indicating that site-specific cloud dynamics should inform model selection in operational forecasting pipelines; the full multi-seed graph-construction ablation and the aleatoric/epistemic decomposition results are in progress and will be reported in full. This work establishes a reproducible GOES-16 benchmark for tropical Andean sites — a climate regime markedly underrepresented in the solar forecasting literature — and contributes one of the first applications of disentangled aleatoric/epistemic uncertainty quantification to satellite-based solar forecasting, supporting robust, risk-aware operational deployment.

1. Introduction

Solar photovoltaic (PV) generation has grown rapidly across Latin America, with Colombia reaching over 1 GW of installed capacity and targeting 9 GW by 2030 under its National Energy Plan [Unidad de Planeación Minero Energética \(UPME\) \(2023\)](#). Reliable short-term forecasts of global horizontal irradiance (GHI) are a prerequisite for efficient operation of PV-integrated power systems: they enable reserve scheduling, reduce imbalance penalties, and support real-time dispatch and grid management. The growing share of variable renewable generation makes this forecasting problem increasingly important: unlike slowly-varying demand, cloud-induced irradiance variability is continuous,

*Corresponding author

 e.ladino@uniandes.edu.co (E.L. Nieto)

ORCID(s):

stochastic, and correlated across spatially distributed generation assets, and accurate, uncertainty-aware short-term forecasts provide operators with the situational awareness needed to schedule reserves, trigger demand response, and avoid costly imbalances.

In tropical countries such as Colombia, the forecasting challenge is compounded by the high frequency and spatial heterogeneity of convective cloud systems, which can reduce surface irradiance by more than 80% within minutes and are poorly captured by coarse-resolution numerical weather prediction (NWP) models. Most published forecasting benchmarks, however, are built around temperate or semi-arid climates, leaving tropical and high-altitude equatorial sites — where cloud dynamics differ substantially — underrepresented in the literature.

Satellite-based nowcasting methods bypass NWP limitations by inferring cloud motion and optical depth directly from geostationary imagery. Instruments such as the GOES-16 Advanced Baseline Imager (ABI) provide multi-spectral observations every 10 minutes at sub-kilometre resolution, making them well-suited for sub-hourly to hourly irradiance estimation and forecasting. The dominant paradigm couples a spatial feature extractor (applied to a patch of satellite pixels centred on the target site) with a temporal model that exploits the recent history of observed GHI or satellite state Zhang, Florita, Hodge, Lu, Hamann, Banunarayanan and Brockway (2015); Reno and Hansen (2016).

Deep learning has substantially advanced this paradigm. Convolutional neural networks (CNNs) applied to satellite patches learn translation-invariant spatial patterns associated with cloud regimes, and when combined with recurrent models such as LSTMs they achieve state-of-the-art performance on benchmark datasets Alessandrini, Monache, Sperati and Cervone (2015); Huertas-Tato, Rodríguez-Benítez, Arbizu-Barrena, Aler-Mur, Galván-León and Pozo-Vázquez (2017); Paletta, Arbod and Lasenby (2021). More recently, graph neural networks (GNNs) have emerged as an alternative spatial encoder: by representing pixels as nodes in a graph and learning neighbourhood aggregation rules, GNNs relax the translation-invariance assumption of CNNs and can capture irregular spatial dependencies introduced by non-uniform cloud structures Hamilton, Ying and Leskovec (2017); Ji, Zhu, Li and Zhou (2022). However, systematic comparisons of GNN-based and CNN-based encoders for GHI forecasting remain scarce, and none, to the best of our knowledge, address the tropical Andean climate.

Contributions

This paper makes the following contributions:

1. We propose **GraphSAGE-LSTM**, a model that treats a GOES-16 satellite patch as a pixel graph, applies GraphSAGE neighbourhood aggregation Hamilton et al. (2017) over it, and passes the resulting sequence of graph embeddings through an LSTM to produce a point forecast of GHI at horizons of 1, 3, and 6 hours. We compare a fixed unweighted nearest-neighbour graph against a learned, distance-weighted k -nearest-neighbour variant with a tunable neighbourhood size, isolating the contribution of *learned spatial connectivity* to forecasting skill.
2. We conduct a **rigorous empirical comparison** across two climatologically distinct Colombian sites, three forecast horizons, and multiple random seeds, including Bayesian hyperparameter optimisation via Optuna Akiba, Sano, Yanase, Ohta and Koyama (2019) for GraphSAGE-LSTM, ResNet-LSTM, and a flat MLP spatial-ablation baseline, together with the graph-construction ablation described above.
3. Preliminary single-seed results indicate that GraphSAGE-LSTM is **competitive with or superior to ResNet-LSTM** among the deep-learning models at multi-hour horizons at both sites, while the absolute skill of both models is markedly lower at the climatologically harder high-altitude site — evidence that site-specific cloud dynamics, not just model architecture, govern attainable forecasting skill. The full multi-seed comparison, including the graph-construction ablation, is in progress (Section 4).
4. We introduce a **disentangled uncertainty-quantification layer** that separately estimates the two sources of forecast uncertainty: an *aleatoric* component, captured by a Gaussian negative-log-likelihood variance head that models irreducible cloud stochasticity, and an *epistemic* component, captured by an ensemble of posterior weight samples obtained via Stochastic Gradient Langevin Dynamics (SGLD) Welling and Teh (2011), which quantifies uncertainty due to limited data and model capacity Kendall and Gal (2017). Unlike a single symmetric prediction interval, this decomposition tells an operator *why* a forecast is uncertain, supporting differentiated, risk-stratified responses (e.g. wider reserve margins for high-epistemic-uncertainty regimes such as under-sampled cloud types, versus accepting irreducible variance during genuinely volatile but well-characterised convective afternoons).
5. We establish a **reproducible GOES-16 benchmark for tropical Andean sites**, including data manifests, patch extraction, model code, Optuna studies, the graph-construction ablation, the SGLD/variance-head uncertainty-decomposition pipeline, and evaluation scripts, addressing the scarcity of published satellite-based GHI

forecasting benchmarks — with disentangled uncertainty quantification — for Latin American and equatorial tropical contexts.

Paper organisation

Section 2 reviews related work. Section 3 describes the datasets, model architectures, training protocol, and evaluation metrics. Section 4 presents quantitative results. Section 5 analyses key findings. Section 6 concludes and outlines future work.

2. Related Work

2.1. Short-term solar irradiance forecasting

Short-term GHI forecasting methods span a wide spectrum from statistical time-series models to physically-based NWP and data-driven machine learning Diagne, David, Lauret, Boland and Schmutz (2013); Voyant, Notton, Kalogirou, Nivet, Paoli, Motte and Fouilloy (2017). For intra-hour and sub-day horizons (0–6 h), satellite-based and ground-observation methods generally outperform NWP, which requires several hours of spin-up time Perez, Kivalov, Schlemmer, Hemker, Renné and Hoff (2010).

Persistence (assuming future irradiance equals the most recent observation) is the standard benchmark for short-term forecasting. It is competitive at very short horizons (under 30 minutes) when irradiance changes slowly, but degrades rapidly beyond one hour Lonij, Brooks, Cronin, Leuthold and Koch (2013).

Statistical models such as ARIMA and its seasonal variant SARIMA have been applied to GHI series after removing the deterministic clear-sky envelope Reikard (2009); Bacher, Madsen and Nielsen (2009). Working on the clear-sky index $k_t = \text{GHI}/\text{GHI}_{\text{cs}}$ improves stationarity and allows standard time-series machinery at hourly resolution. Despite their simplicity, seasonal ARIMA models can be competitive at multi-hour horizons in climates with regular diurnal patterns Pedro and Coimbra (2012).

2.2. Deep learning with satellite imagery

CNN-based architectures applied to sequences of geostationary satellite patches have become the dominant approach for satellite-based GHI forecasting. Early work showed that cloud-motion vectors extracted from visible channels improve short-term forecasts Chow, Urquhart, Lave, Dominguez, Kleissl, Shields and Washom (2011). More recent studies feed raw satellite reflectances directly into CNNs followed by fully-connected regressors or LSTMs Huertas-Tato et al. (2017); Paletta et al. (2021); Feng, Yang, Su, Qin, Yi, Zheng and Zhen (2022). The encoder-decoder design, where a CNN extracts per-frame spatial embeddings that are then processed by a recurrent model, has shown consistent gains over purely spatial or purely temporal baselines Wang, Mi, Li and Zhao (2019).

Attention-based architectures including Vision Transformers (ViT) and Swin Transformers have also been applied to satellite-based solar forecasting Acıkgöz (2022); Liu, Yang, Wang and Wang (2023), achieving competitive performance at the cost of increased parameter count and data requirements. For moderate dataset sizes such as those encountered in single-site studies, CNN+LSTM hybrids remain a practical and well-characterised baseline.

2.3. Graph neural networks for spatial data

GraphSAGE Hamilton et al. (2017) is an inductive GNN that learns aggregation functions over local neighbourhoods, enabling generalisation to unseen nodes. It has been applied to traffic forecasting Li, Yu, Shahabi and Liu (2018), air quality prediction, and wind power generation Kazem, Chaichan, Al-Waeli and Sopian (2021), where spatial proximity defines the graph topology. In the context of satellite imagery, representing pixels as graph nodes allows the model to capture irregular spatial dependencies that periodic CNN filters may miss, such as cloud boundaries or fractional cloud cover patterns.

Closest to our work, Ji et al. (2022) applied a GNN-LSTM model to multi-site solar irradiance forecasting, treating monitoring stations as nodes. Our approach differs in that we build the graph within a single satellite patch (pixel level), without requiring multi-station measurements, which is more practical for single-site deployment.

2.4. Forecasting in tropical and Andean climates

Most GHI forecasting benchmarks are conducted in temperate or semi-arid climates (Europe, southwestern United States, Australia) Paletta, Arbod and Lasenby (2022); Yang, Kleissl, Gueymard, Pedro and Coimbra (2018). Tropical Andean sites present distinct challenges: convective cloud formation is driven by topographic lifting and moisture advection, producing strong intraday variability concentrated in the afternoon Poveda, Jaramillo and Vallejo (2014).

Table 1

Study site metadata.

	El Paso (César)	Uniandes (Bogotá)
Latitude	9.737°N	4.602°N
Longitude	73.695°W	74.066°W
Elevation	50 m a.s.l.	2 600 m a.s.l.
Climate	Tropical semi-arid	Equatorial mountain
Avg. GHI	~5.5 kWh/m ² /day	~4.2 kWh/m ² /day

Few published studies address satellite-based forecasting for Colombian or equivalent tropical mountain conditions, motivating the present work.

2.5. Forecasting for power system operation and decision-making

Accurate short-term irradiance forecasts feed directly into operational decision-making for solar-integrated power systems: they support reserve scheduling, unit commitment, and demand-response triggers, reducing the cost of accommodating variable renewable generation Feng, Zhang, Sun and Hodge (2021). For renewable integration specifically, accurate forecasting reduces the need for spinning reserves and allows pre-emptive load balancing, directly limiting the impact of generation uncertainty on grid stability.

Beyond point forecasts, operators benefit from knowing *why* a forecast is uncertain. A forecast can be uncertain because the underlying cloud field is intrinsically stochastic and would remain so even with infinite training data (*aleatoric* uncertainty), or because the model has seen too little data resembling the current conditions and would become more confident with more observations (*epistemic* uncertainty) Kendall and Gal (2017). The two sources call for different operational responses — accepting irreducible variance during well-characterised but volatile conditions versus provisioning extra reserve margin under genuinely novel, under-sampled regimes — a distinction that a single symmetric or distribution-free interval cannot make on its own.

2.6. Uncertainty quantification in deep learning for forecasting

Bayesian deep learning provides a principled route to decomposing predictive uncertainty. Kendall and Gal (2017) formalise the aleatoric/epistemic split for deep regression and segmentation, combining a heteroscedastic Gaussian negative-log-likelihood (NLL) head for aleatoric noise with Monte Carlo sampling over model weights for epistemic uncertainty. Approximate posterior sampling over weights can be obtained at training-time cost via Stochastic Gradient Langevin Dynamics (SGLD, Welling and Teh (2011)), which injects calibrated Gaussian noise into stochastic gradient updates so that, under a decaying step size, the resulting trajectory asymptotically samples the true posterior $p(\theta \mid \mathcal{D})$; collecting weight snapshots along a single SGLD chain yields an inexpensive approximate ensemble whose spread quantifies epistemic uncertainty, distinct from — and complementary to — the aleatoric component captured by a heteroscedastic variance head. These ideas have seen limited application to satellite-based solar forecasting, where the dominant uncertainty driver — stochastic convective cloud formation — is a textbook aleatoric source that is rarely modelled explicitly alongside the model's own epistemic limitations.

This work contributes to this literature by (i) developing spatially-aware deep learning for solar forecasting at tropical Andean sites that remain underrepresented in existing benchmarks, and (ii) applying an aleatoric/epistemic uncertainty decomposition — via a variance head and an SGLD posterior ensemble — to satellite-based GHI forecasting, supporting risk-stratified, robust operational decisions that a single point forecast or symmetric interval cannot provide.

3. Methodology

3.1. Study sites

Experiments are conducted at two meteorological stations in Colombia that represent climatologically contrasting regimes (Table 1).

El Paso is located in the lowland Caribbean region of César department, characterised by high annual irradiance, predominantly clear-sky conditions, and dry-season cloud cover driven by the Inter-Tropical Convergence Zone (ITCZ).

Universidad de los Andes (referred to as Uniandes) is an instrumented rooftop station in the Colombian Andean city of Bogotá. At 2 600 m above sea level, it experiences the equatorial bimodal rainfall regime with intense convective cloud formation typically peaking in the early afternoon, producing sharp intraday GHI drops.

3.2. Data

3.2.1. Ground measurements

GHI is measured at 10-minute resolution and stored in time-zone-aware (UTC) Parquet files. Missing values account for less than 2% of the record at both sites and are excluded during dataset construction.

3.2.2. Satellite data and patch extraction

We use the GOES-16 ABI Level-2 Multi-Band Cloud and Moisture Imagery Product (ABI-L2-MCMIPF), which provides all 16 ABI spectral channels at a unified 10-minute cadence and ~ 2 km spatial resolution in the visible bands. Raw hourly ABI-L2-MCMIPF granules are pre-processed into compact per-hour archives: each archive is a compressed NumPy array of shape $(6, 16, P, P)$, holding the six 10-minute slots within that hour, the $C = 16$ spectral channels, and a $P \times P = 16 \times 16$ pixel patch ($P = 16$) centred on the target site, in units consistent with each band's native reflectance or brightness temperature. At inference/training time, a single timestamp is resolved to its $(C, P, P) = (16, 16, 16)$ frame by selecting the corresponding hourly archive and 10-minute slot, so no redundant decoding of the original satellite granules is required.

3.2.3. Manifests, sequence construction, and temporal split

For the deep-learning models (ResNet-LSTM, GraphSAGE-LSTM, FlatMLP), forecast samples are constructed as pairs (X, y) where the input $X \in \mathbb{R}^{L \times C \times P \times P}$ is a sequence of $L = 24$ satellite patch frames covering the most recent 4 hours of observation (10-minute cadence), and the target $y \in \mathbb{R}$ is the ground-measured GHI at the site H steps ahead ($H \in \{6, 18, 36\}$ steps, corresponding to 1, 3, and 6 hours).

A pre-computed **manifest** (one Parquet table per site and horizon) enumerates every valid forecast sample: for each candidate target timestamp it stores the label y , the list of $L = 24$ history timestamps `history_ts`, and a coverage flag confirming that (i) all L history steps exist in the ground-truth series and (ii) a satellite archive is available for each of them. Samples for which any of the L history frames is missing on disk are filtered out at load time, so every batch the models see contains a complete 4-hour satellite history. Two thin Dataset wrappers consume the same manifest and patch archives but reshape the retrieved frames differently for the two model families:

- `PatchSeqDataset` (used by ResNet-LSTM and FlatMLP) returns each sample as an image-like sequence $X \in \mathbb{R}^{L \times C \times P \times P}$ — i.e., L stacked $(16, 16, 16)$ tensors preserving the 2-D pixel grid, ready to be consumed by 2-D convolutions.
- `GraphSeqDataset` (used by GraphSAGE-LSTM) returns each sample as a node-feature sequence $X \in \mathbb{R}^{L \times N \times C}$ with $N = P^2 = 256$: every frame is transposed and reshaped from (C, P, P) to (P^2, C) , so that each of the 256 pixels becomes a graph node carrying a $C = 16$ -dimensional feature vector (its 16-channel spectral signature at that location).

To prevent data leakage, train, validation, and test sets are defined by non-overlapping calendar blocks (Table 2), and the manifest coverage check, splitting boundaries, and patch archives are shared across all three deep-learning models, ensuring that ResNet-LSTM, GraphSAGE-LSTM, and FlatMLP are trained and evaluated on *exactly* the same samples.

3.3. Models

3.3.1. Persistence baseline

The persistence model assumes irradiance is unchanged over the forecast horizon: $\hat{y}(t + H) = \text{GHI}(t)$. It is optimal under no-change dynamics and serves as the standard benchmark for short-term solar forecasting.

3.3.2. SARIMA: classical statistical baseline

As a non-deep-learning reference, we fit a seasonal ARIMA model, $\text{SARIMAX}(2, 1, 2)(1, 1, 1)_{24}$, directly on the raw hourly GHI series at each site (order selected by grid search on validation AIC; no clear-sky normalisation is applied, so the model targets the same physical quantity, GHI in W/m^2 , as every other model). Because a seasonal

Table 2

Temporal data splits.

Site	Split	Start	End	n (approx.)
El Paso	Train	2022-03-01	2023-06-30	69 984
	Val	2023-07-01	2023-10-31	17 568
	Test	2023-11-01	2024-03-07	18 288
Unianandes	Train	2023-09-01	2024-09-30	56 850
	Val	2024-10-01	2024-12-31	13 104
	Test	2025-01-01	2025-03-28	12 384

period of $m = 144$ (10-minute native resolution) is computationally intractable over the multi-year training window, the satellite-resolution series is first resampled to hourly cadence ($m = 24$), and SARIMA is therefore evaluated only on the subset of test timestamps that fall on an exact hour, with the persistence reference *recomputed on that same coarser subset* so that SARIMA's own skill score is internally consistent. **This makes SARIMA's RMSE_{day} and Skill_{day} not directly comparable to the full 10-minute-resolution metrics reported for persistence and the deep-learning models elsewhere in this paper;** it is included in Figs. 1–2 as a classical-statistical reference point, visually flagged (hatched bars), rather than in Table 5.

3.3.3. ResNet-LSTM: convolutional spatial encoder

ResNet-LSTM is an encoder–decoder that treats each satellite frame as a conventional multi-channel image and lets a convolutional network learn its spatial representation, before a recurrent network reasons over the sequence of representations.

Spatial encoder (per frame). Each frame $X_\ell \in \mathbb{R}^{C \times P \times P}$ ($\ell = 1, \dots, L$) is passed independently through a lightweight ResNet He, Zhang, Ren and Sun (2016): a 3×3 convolutional stem, followed by three residual stages of BasicBlock units (two 3×3 convolutions with batch normalisation and a skip connection). The first stage preserves the 16×16 spatial resolution, while the second and third stages apply stride-2 downsampling, taking the spatial grid through $16 \rightarrow 8 \rightarrow 4$. A global-average-pooling layer then collapses the 4×4 feature map to a single vector, which a final linear layer projects to a d_e -dimensional frame embedding $\mathbf{e}_\ell \in \mathbb{R}^{d_e}$ (emb_dim, tuned in $\{64, 128, 192, 256\}$). The encoder's channel width is controlled by base $\in \{16, 24, 32, 48\}$ (the three stages use base, $2 \times$ base, and $4 \times$ base channels respectively); the resulting parameter count ranges from approximately 50k to 500k depending on the configuration. Because the same encoder weights are applied to every one of the $L = 24$ frames, the model is translation-equivariant and shares a single spatial template across the whole 4-hour history.

Temporal encoder. The L frame embeddings are normalised (LayerNorm) and stacked into a sequence $(\mathbf{e}_1, \dots, \mathbf{e}_L) \in \mathbb{R}^{L \times d_e}$, which a single- or two-layer LSTM Hochreiter and Schmidhuber (1997) (hidden_t, n_lstm_layers) processes step by step; only its final hidden state $\mathbf{h}_L \in \mathbb{R}^{d_t}$ is retained. A two-layer MLP head (Linear $\rightarrow$ ReLU $\rightarrow$ Dropout $\rightarrow$ Linear) maps $\mathbf{h}_L$ to the scalar point forecast $\hat{y}$.

3.3.4. GraphSAGE-LSTM: graph-based spatial encoder

GraphSAGE-LSTM shares the same encoder–decoder skeleton as ResNet-LSTM — spatial encoder per frame, then LSTM over the embedding sequence, then MLP head — but *replaces the convolutional encoder with a graph neural network* that treats individual pixels, rather than the whole frame, as the basic spatial unit.

From pixels to a graph. Each (C, P, P) frame is reshaped (by GraphSeqDataset, Section 3.2.3) into $N = P^2 = 256$ nodes, one per pixel, each carrying its native $C = 16$ -channel spectral vector as a node feature — no convolution or pooling is applied beforehand. A graph $G = (V, E, \mathbf{w})$ is built once over the 16×16 pixel grid and reused for every frame and every sample; the edge index and weights are precomputed once, registered as model buffers, and replicated across the batch dimension, so the same mini-batch DataLoader and training loop used for ResNet-LSTM apply unchanged. We compare two graph-construction variants, isolating the contribution of *learned spatial connectivity* to forecasting skill (ablation reported in Section 4):

- **Fixed unweighted graph (baseline).** Each node connects to its 8 immediate grid neighbours (cardinal and diagonal) with unit edge weight, the standard local-connectivity topology used in early graph convolutional networks applied to image-like data.

- **Learned weighted k -NN graph (proposed).** Each node connects to its k closest pixels by Euclidean distance in 2-D pixel coordinates ($k \in \{4, 8, 12, 16\}$, tuned per site/horizon, Section 3.4), and every edge is assigned an inverse-distance weight $w_{vu} = 1/d(v, u)$ (Eq. 1 below). Larger k values progressively add diagonal, two-step, and knight-move neighbours, so k effectively controls the spatial receptive field of a single graph layer. Unlike the fixed baseline or a convolutional filter, this topology lets the model learn *different* aggregation weights for differently-located pixels (e.g. patch centre vs. periphery), rather than sliding one shared template across the image, and lets Optuna select the neighbourhood size that best matches each site’s cloud spatial scale.

$$w_{vu} = \frac{1}{d(v, u)}, \quad (v, u) \in E. \quad (1)$$

Spatial encoder (per frame). GraphSAGE Hamilton et al. (2017) updates every node’s representation by aggregating its neighbourhood. With edge weights, the neighbourhood mean becomes a distance-weighted average, and the update concatenates each node’s own (self) representation with that weighted-neighbour summary before a shared linear projection and ReLU:

$$\mathbf{h}_v^{(k)} = \sigma \left(\mathbf{W}^{(k)} \left[\mathbf{h}_v^{(k-1)} \parallel \frac{\sum_{u \in \mathcal{N}(v)} w_{vu} \mathbf{h}_u^{(k-1)}}{\sum_{u \in \mathcal{N}(v)} w_{vu}} \right] \right) \quad (2)$$

where $\mathcal{N}(v)$ denotes the neighbours of v under whichever graph variant is in use, $\parallel$ is concatenation, and σ is ReLU. For the fixed unweighted baseline, $w_{vu} = 1$ for all $(v, u) \in E$, reducing Eq. 2 to the standard unweighted SAGE neighbourhood mean; the two graph variants therefore share the same aggregation layer and differ only in E and $\mathbf{w}$, isolating the effect of graph construction from the rest of the architecture. The same layer (with shared weights $\mathbf{W}^{(k)}$, output width `hidden_g`) is applied to all 256 nodes simultaneously and stacked $K \in \{2, 3, 4\}$ times (`n_sage_layers`); each additional layer extends a node’s effective receptive field by one more k -NN hop. A final *mean readout* over all $N = 256$ nodes collapses the per-pixel representations into a single graph-level (frame) embedding $\mathbf{g}_\ell \in \mathbb{R}^{d_g}$, with $d_g = \text{hidden_g} \in \{64, 96, 128, 192, 256\}$ — the GraphSAGE analogue of the ResNet’s d_e -dimensional frame embedding.

Temporal encoder. As in ResNet-LSTM, the L graph-level embeddings are normalised and stacked into $(\mathbf{g}_1, \dots, \mathbf{g}_L) \in \mathbb{R}^{L \times d_g}$ and processed by a single- or two-layer LSTM whose final hidden state feeds an identical MLP head producing $\hat{y}$. The two architectures are thus directly comparable: they differ *only* in how the per-frame spatial embedding is produced — convolution-and-downsampling over a regular grid (ResNet) versus weighted message-passing over a learned k -NN pixel graph at full 256-node resolution, with no spatial downsampling (GraphSAGE) — while sharing the same temporal model, output head, training protocol, and data pipeline.

3.3.5. FlatMLP: temporal-only ablation

To isolate the contribution of *spatial* encoding from that of temporal modelling, FlatMLP discards all pixel-level spatial structure. For every frame $X_\ell \in \mathbb{R}^{C \times P \times P}$, a global spatial average pool collapses the $P \times P$ grid to its per-channel mean, $\bar{\mathbf{x}}_\ell \in \mathbb{R}^C$ — i.e. each frame is reduced to a 16-dimensional “patch-mean spectral signature” with no notion of pixel location. The $L = 24$ resulting vectors are concatenated into a single flat input $\bar{\mathbf{x}} \in \mathbb{R}^{L \times C} = \mathbb{R}^{384}$ (24×16). This vector is passed through a fully-connected feed-forward network of `n_layers` blocks, each consisting of `Linear` $\rightarrow$ `LayerNorm` $\rightarrow$ `ReLU` $\rightarrow$ `Dropout` (`hidden_dim` units), followed by a final `Linear` layer that outputs the scalar $\hat{y}$ directly — every unit in one layer is connected to every unit in the next (a standard dense MLP), with no recurrence, no convolution, and no graph structure. Because FlatMLP receives exactly the same $L \times C \times P \times P$ input tensors (via `PatchSeqDataset`) and is trained with the same protocol, its skill gap relative to ResNet-LSTM and GraphSAGE-LSTM directly quantifies how much of their forecasting skill stems from *spatial* encoding of the satellite patch, as opposed to simple per-frame statistics processed over time.

3.4. Hyperparameter optimisation

For the three deep-learning models (ResNet-LSTM, GraphSAGE-LSTM, and FlatMLP) we use Optuna Akiba et al. (2019) with the TPE sampler and Median pruner. The fixed-graph GraphSAGE-LSTM baseline, the corresponding ResNet-LSTM configuration, and FlatMLP use a budget of 50 trials per configuration (site $\times$ horizon $\times$ seed) across

Table 3

Optuna hyperparameter search spaces. L1 coefficient and LSTM depth are new relative to the initial baseline study. The `k_neighbors` parameter determines the weighted k -NN graph built per trial for GraphSAGE-LSTM.

Hyperparameter	ResNet-LSTM	GraphSAGE-LSTM
<code>base / hidden_g</code>	{16, 24, 32, 48}	{64, 96, 128, 192, 256}
<code>emb_dim / n_sage_layers</code>	{64, 128, 192, 256}	{2, 3, 4}
<code>hidden_t (LSTM)</code>	{64, 128, 192, 256}	{64, 96, 128, 192, 256}
<code>n_lstm_layers</code>	{1, 2}	{1, 2}
<code>k_neighbors</code>	—	{4, 8, 12, 16}
<code>l1_reg</code>	{0, 10^{-5} , 10^{-4} , 10^{-3} }	{0, 10^{-5} , 10^{-4} , 10^{-3} }
<code>dropout</code>	{0, 0.1, 0.2, 0.3}	{0, 0.1, 0.2}
<code>lr</code>	[3×10^{-4} , 3×10^{-3}]	[5×10^{-5} , 2×10^{-3}]
<code>weight_decay</code>	[10^{-6} , 10^{-3}]	[10^{-6} , 10^{-3}]
<code>batch_size</code>	{16, 32, 64}	{8, 16, 32}

four random seeds {42, 1, 7, 13}. The extended search space used for the weighted k -NN ablation (Table 3) increases per-trial cost substantially — larger embeddings, optionally deeper LSTMs, and a tunable graph rebuilt per trial — so this variant uses a reduced budget of 75 trials across two random seeds {42, 1}, prioritising coverage of the most consequential hyperparameters (graph connectivity, model capacity) over seed count; results are reported with this caveat wherever the two protocols are compared (Section 4).

The ResNet-LSTM and GraphSAGE-LSTM search spaces (Table 3) include `n_lstm_layers` $\in$ {1, 2} to allow stacked recurrent processing, and a categorical L1 regularisation coefficient (`l1_reg`) to penalise large weights in high-capacity configurations; both are new relative to the initial baseline study. For **ResNet-LSTM** specifically, the embedding dimension (`emb_dim`) and temporal hidden size (`hidden_t`) are extended to 256 to increase capacity. For **GraphSAGE-LSTM**, the number of nearest neighbours ($k \in$ {4, 8, 12, 16}) used to build the weighted pixel graph is additionally tunable, allowing the study to determine the optimal spatial connectivity for each site-horizon combination. The fixed-graph baseline (Section 3.3.4) is tuned with the same search space, excluding `k_neighbors` since its topology is fixed a priori, and serves as the ablation reference for the value of learned spatial connectivity. For **FlatMLP**, the study searches over MLP depth, hidden dimension, dropout, learning rate, and weight decay (50 trials).

The Optuna study identifies the best trial on validation RMSE_{day}; the winning configuration is then retrained for up to 30 epochs with patience 8 and evaluated on the held-out test set.

3.5. Training protocol

All deep-learning models are trained with the Adam optimiser minimising the MSE loss on normalised targets ($y_{\text{norm}} = (y - \bar{y})/s_y$, with $\bar{y}$ and s_y estimated from the training set). Training uses early stopping with patience 6 (trials) or 8 (final retraining). Automatic Mixed Precision (AMP) is enabled when a GPU is available. Each experiment is replicated with four seeds {42, 1, 7, 13} to report mean $\pm$ std statistics, except the weighted k -NN ablation, which uses two seeds {42, 1} as noted in Section 3.4.

3.6. Uncertainty quantification: disentangling aleatoric and epistemic uncertainty

The best-performing backbone identified in Sections 3.3.3–3.3.5 is extended into a probabilistic model that separately estimates the two sources of forecast uncertainty Kendall and Gal (2017): *aleatoric* uncertainty, the irreducible stochasticity of cloud formation that would remain even with infinite training data, and *epistemic* uncertainty, the reducible uncertainty due to limited training data and finite model capacity. Reporting a single undifferentiated interval cannot distinguish a forecast that is uncertain because the cloud field is genuinely unpredictable from one that is uncertain because the model has not seen similar conditions before; the two call for different operational responses (Section 1).

Aleatoric uncertainty: heteroscedastic Gaussian head. The deterministic scalar output head (Sections 3.3.3–3.3.5) is replaced with a two-head output ($\mu_\theta(x), \log \sigma_\theta^2(x)$) predicting, respectively, the conditional mean and log-variance of $y | x$, trained by minimising the Gaussian negative log-likelihood

$$\mathcal{L}_{\text{NLL}}(\theta) = \frac{1}{2\sigma_\theta^2(x)} (y - \mu_\theta(x))^2 + \frac{1}{2} \log \sigma_\theta^2(x), \quad (3)$$

which lets the model assign larger predicted variance to inputs it judges intrinsically noisier (e.g. rapidly evolving convective afternoons), penalised by the log-variance term against trivially inflating σ^2 everywhere.

Epistemic uncertainty: SGLD posterior sampling. Standard SGD/Adam training returns a single point estimate $\hat{\theta}$ of the network weights. To approximate the posterior distribution $p(\theta \mid \mathcal{D})$ instead, we use Stochastic Gradient Langevin Dynamics (SGLD, Welling and Teh (2011)), which injects calibrated Gaussian noise into each gradient step:

$$\theta_{t+1} = \theta_t - \epsilon_t \nabla_{\theta} \mathcal{L}(\theta_t) + \sqrt{2\epsilon_t} \eta_t, \quad \eta_t \sim \mathcal{N}(0, I), \quad (4)$$

where ϵ_t is the (decaying) step size. Under suitable conditions on ϵ_t , the resulting Markov chain asymptotically samples from $p(\theta \mid \mathcal{D})$ without the computational overhead of full MCMC. Following a burn-in period of $T_{\text{burn-in}}$ epochs (discarded, while the chain moves away from the SGD initialisation), we collect T weight snapshots $\{\theta_t\}_{t=1}^T$ at a fixed sampling interval Δ (epochs), yielding an inexpensive approximate ensemble.

Combining both into a total-uncertainty decomposition. For a test input x , each posterior sample θ_t produces a pair $(\mu_t(x), \sigma_t^2(x))$ via the NLL head. The ensemble mean $\bar{\mu}(x) = \frac{1}{T} \sum_{t=1}^T \mu_t(x)$ is the point forecast, and the total predictive variance decomposes, by the law of total variance, into an aleatoric and an epistemic term Kendall and Gal (2017):

$$\underbrace{\frac{1}{T} \sum_{t=1}^T (\mu_t(x) - \bar{\mu}(x))^2}_{\text{epistemic } \sigma_{\text{epi}}^2(x)} + \underbrace{\frac{1}{T} \sum_{t=1}^T \sigma_t^2(x)}_{\text{aleatoric } \sigma_{\text{ale}}^2(x)} = \sigma_{\text{total}}^2(x). \quad (5)$$

$\sigma_{\text{epi}}^2(x)$ is large where the T posterior samples disagree on $\mu_t(x)$ — i.e. where training data was scarce or atypical — and shrinks as more representative training data becomes available, even without retraining the variance head. $\sigma_{\text{ale}}^2(x)$ instead reflects the model’s learned, input-dependent estimate of irreducible noise and would not shrink with more data. We report both components separately, alongside the combined predictive interval $\bar{\mu}(x) \pm z_{1-\alpha/2} \sigma_{\text{total}}(x)$, and evaluate empirical coverage at the same nominal levels used in prior versions of this pipeline ($1 - \alpha \in \{80\%, 90\%, 95\%\}$) for comparability, in addition to reporting the aleatoric/epistemic split itself as the primary object of interest.

3.7. Evaluation metrics

Following standard practice in solar forecasting Coimbra, Kleissl and Marquez (2013), we report:

Daytime RMSE (RMSE_{day} , W/m^2): RMSE computed only over test samples with observed GHI $\geq 20 \text{ W/m}^2$, excluding night-time samples where all models trivially predict zero.

Daytime skill score ($\text{Skill}_{\text{day}}$): normalised improvement over persistence,

$$\text{Skill}_{\text{day}} = 1 - \frac{\text{RMSE}_{\text{day}}(\text{model})}{\text{RMSE}_{\text{day}}(\text{persistence})} \quad (6)$$

A score of 0 indicates parity with persistence; positive values indicate improvement. We also report MAE_{day} for completeness.

4. Results

4.1. Persistence reference

Table 4 reports the persistence RMSE_{day} at each site and horizon, which defines the denominator of $\text{Skill}_{\text{day}}$ for all models.

The 1-hour persistence error at El Paso (204.4 W/m^2) is notably lower than at Uniandes (293.9 W/m^2), reflecting El Paso’s more stable clear-sky regime where short-horizon irradiance changes are small. At 6 hours, both sites have comparable absolute difficulty (565.3 vs. 470.9 W/m^2).

4.2. Main results

Table 5 reports RMSE_{day} and $\text{Skill}_{\text{day}}$ for all models on the test set. Deep-learning results are mean $\pm$ std over four seeds.

Table 4

Persistence test-set RMSE_{day} (W/m^2). All deep-learning models are compared against these values.

Site	1h	3h	6h
El Paso	204.4	411.0	565.3
Uniandes	293.9	405.0	470.9

Table 5

Test-set results. RMSE_{day} in W/m^2 . $\text{Skill}_{\text{day}}$ relative to persistence (higher is better). DL models: mean $\pm$ std over 4 seeds. Best $\text{Skill}_{\text{day}}$ per site-horizon highlighted in **bold**.

Model	Metric	El Paso (César)			Uniandes (Bogotá)		
		1h	3h	6h	1h	3h	6h
Persistence	RMSE_{day}	204.4	411.0	565.3	293.9	405.0	470.9
	$\text{Skill}_{\text{day}}$	0.000	0.000	0.000	0.000	0.000	0.000
ResNet-LSTM	RMSE_{day}	238.6 ± 22.4	238.5 ± 42.7	218.4 ± 13.2	246.7 ± 4.1	255.7 ± 4.8	297.9 ± 8.8
	$\text{Skill}_{\text{day}}$	-0.167 ± 0.110	$+0.420 \pm 0.104$	$+0.614 \pm 0.023$	$+0.161 \pm 0.014$	$+0.369 \pm 0.012$	$+0.368 \pm 0.019$
GraphSAGE-LSTM	RMSE_{day}	244.7 ± 8.1	238.9 ± 15.9	206.1 ± 7.4	250.8 ± 5.1	252.5 ± 1.5	274.4 ± 3.9
	$\text{Skill}_{\text{day}}$	-0.197 ± 0.040	$+0.419 \pm 0.039$	$+0.636 \pm 0.013$	$+0.147 \pm 0.017$	$+0.376 \pm 0.004$	$+0.417 \pm 0.008$
FlatMLP (Optuna)	RMSE_{day}	—	—	—	—	—	—
	$\text{Skill}_{\text{day}}$	—	—	—	—	—	—

GraphSAGE-LSTM rows use the **fixed unweighted graph** variant (Section 3.3.4); results for the learned weighted k -NN variant are in progress and reported separately in Table 6. FlatMLP (Optuna) results are pending completion of the four-seed sweep (Section 5, Limitations) and will be populated prior to submission; this row quantifies what fraction of deep-learning skill arises from spatial encoding versus temporal dynamics alone.

Table 6

Graph-construction ablation for GraphSAGE-LSTM. $\text{Skill}_{\text{day}}$ relative to persistence (higher is better). Fixed graph: mean $\pm$ std over 4 seeds. Weighted k -NN graph: mean $\pm$ std over 2 seeds {42, 1} (Section 3.4). [Weighted k -NN ablation in progress; values to be populated as runs complete.]

Graph variant	Metric	El Paso (César)			Uniandes (Bogotá)		
		1h	3h	6h	1h	3h	6h
Fixed unweighted, $k = 8$ (4 seeds)	$\text{Skill}_{\text{day}}$	-0.197 ± 0.040	$+0.419 \pm 0.039$	$+0.636 \pm 0.013$	$+0.147 \pm 0.017$	$+0.376 \pm 0.004$	$+0.417 \pm 0.008$
Weighted k -NN, tuned (2 seeds)	$\text{Skill}_{\text{day}}$	—	—	—	—	—	—
	best k	—	—	—	—	—	—

4.3. Graph-construction ablation: fixed vs. learned spatial connectivity

Table 6 isolates the contribution of *learned spatial connectivity* by comparing the fixed unweighted 8-neighbour graph (used for the GraphSAGE-LSTM rows of Table 5) against the learned, distance-weighted k -nearest-neighbour graph with Optuna-tuned neighbourhood size k (Section 3.3.4), holding the rest of the architecture, training protocol, and data pipeline fixed. The weighted-graph variant uses a reduced two-seed protocol (Section 3.4); we report it alongside the four-seed fixed-graph baseline with this caveat rather than withholding it pending a full four-seed re-run.

4.4. Horizon-by-horizon analysis

1-hour horizon. Both deep-learning models fail to beat persistence at El Paso: ResNet-LSTM ($\text{Skill}_{\text{day}} = -0.167 \pm 0.110$) and GraphSAGE-LSTM (-0.197 ± 0.040) show negative skill, meaning they produce higher RMSE than simply repeating the last observation. At Uniandes the picture is more favourable: ResNet-LSTM achieves modest positive skill ($+0.161 \pm 0.014$), narrowly ahead of GraphSAGE-LSTM ($+0.147 \pm 0.017$). The key difference between sites is the persistence reference: El Paso's 1-hour persistence RMSE (204.4 W/m^2) is comparatively low because irradiance changes slowly under predominantly clear-sky conditions, leaving little room for improvement from data-driven methods. Uniandes' 1-hour persistence RMSE (293.9 W/m^2) is higher due to the frequent cloud passages, which create more structure for the models to exploit.

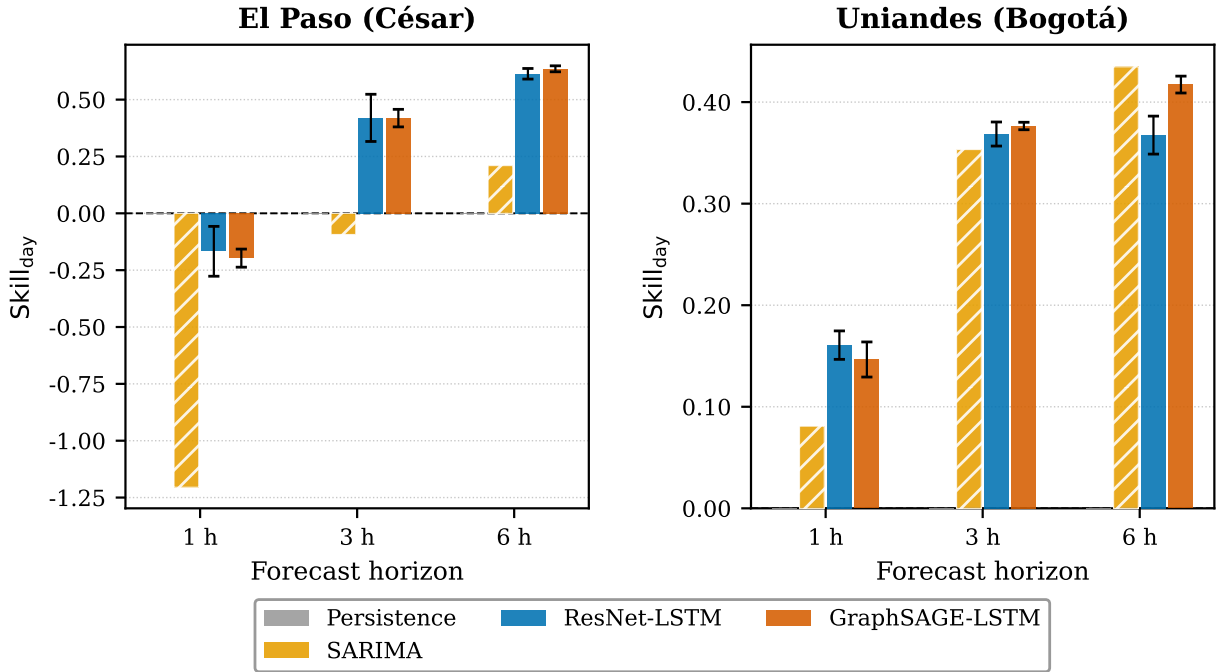


Figure 1: Daytime skill score ($\text{Skill}_{\text{day}}$) for ResNet-LSTM, GraphSAGE-LSTM (both Optuna-tuned, fixed-graph variant, four seeds), persistence, and SARIMA, at both sites across three forecast horizons. Bars show mean over four seeds; error bars indicate ± 1 standard deviation. Persistence has no seed variability. The hatched SARIMA bars are evaluated on a coarser, on-the-hour subset of the test set with their own recomputed persistence reference (Section 3.3.2) and are **not directly comparable** to the other bars; FlatMLP is omitted pending completion of its Optuna sweep (Section 5, Limitations). GraphSAGE-LSTM achieves the highest skill among the deep-learning models at the 6-hour horizon at both sites (0.636 at El Paso; 0.417 at Uniandes).

3-hour horizon. Both deep-learning models outperform persistence at El Paso by approximately 42% ($\text{Skill}_{\text{day}} \approx 0.42$ for both architectures, with GraphSAGE-LSTM marginally ahead at $+0.419 \pm 0.039$). The large standard deviation for ResNet-LSTM (± 0.090) reflects seed-to-seed variability, with some seeds achieving near-GraphSAGE performance and others considerably below.

At Uniandes, GraphSAGE-LSTM attains the best skill at this horizon ($+0.376 \pm 0.004$), narrowly ahead of ResNet-LSTM ($+0.369 \pm 0.012$); both are similar to each other and slightly below their El Paso counterparts despite the higher absolute persistence error at this site.

6-hour horizon. This is the most favourable horizon for the deep-learning models. **GraphSAGE-LSTM achieves the best overall result:** $\text{RMSE}_{\text{day}} = 206.1 \pm 7.4 \text{ W/m}^2$ and $\text{Skill}_{\text{day}} = 0.636 \pm 0.013$ at El Paso, reducing the persistence RMSE by 63.6%. ResNet-LSTM is close (0.614 ± 0.023) but consistently below GraphSAGE across all seeds.

At Uniandes, GraphSAGE-LSTM again leads (0.417 ± 0.008 vs. 0.368 ± 0.019 for ResNet-LSTM), with a wider margin than at El Paso ($+0.049$ vs. $+0.022$ percentage points of skill). However, the *absolute* skill at Uniandes remains markedly lower than at El Paso for both architectures — the gap we analyse in Section 5.

4.5. Statistical robustness

The standard deviation across seeds provides a measure of training stability. GraphSAGE-LSTM consistently shows lower seed-to-seed variability than ResNet-LSTM (e.g., El Paso 6h: ± 0.013 vs. ± 0.023), suggesting that the graph structure regularises training and reduces sensitivity to weight initialisation. The most unstable model is ResNet-LSTM at El Paso 3h (± 0.090), which may reflect the interaction between the high-irradiance regime and the CNN's sensitivity to specific initialisation-driven features.

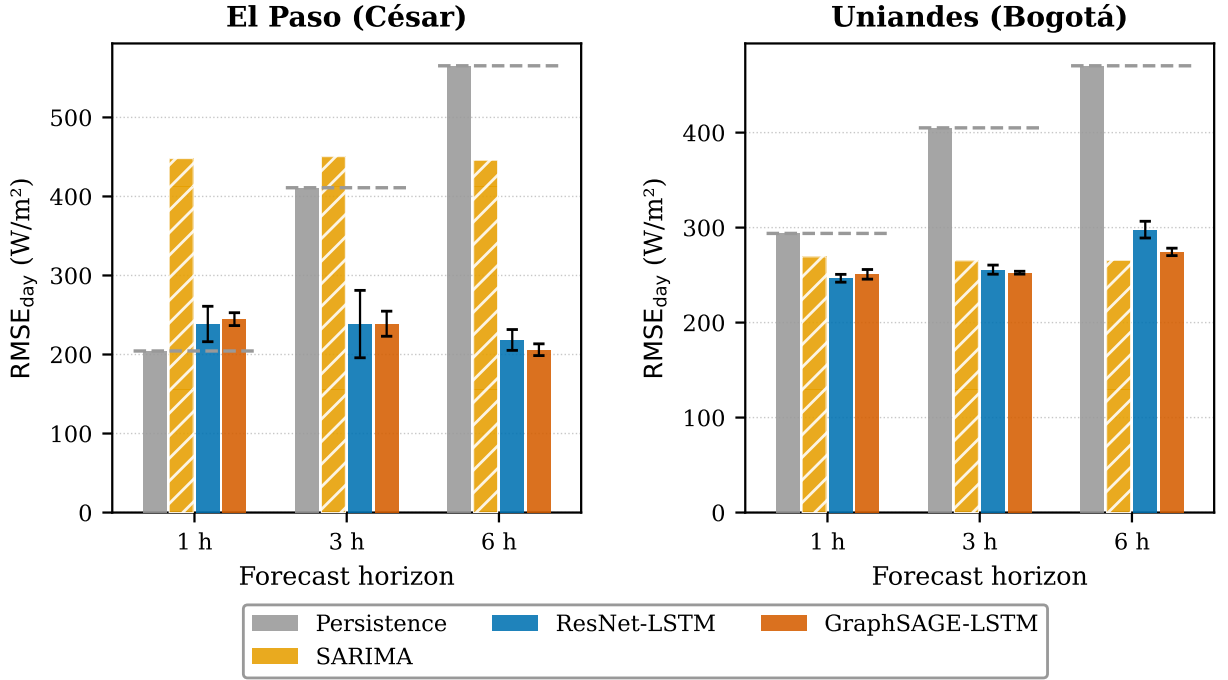


Figure 2: Daytime RMSE (RMSE_{day} , W/m^2) for ResNet-LSTM, GraphSAGE-LSTM, persistence, and SARIMA. Dashed horizontal lines show the persistence reference for each horizon group. The hatched SARIMA bars use a coarser on-the-hour test subset and are **not directly comparable** to the other bars (Section 3.3.2); FlatMLP is omitted pending completion of its Optuna sweep. Both Optuna-tuned deep-learning models substantially reduce RMSE relative to persistence at 3h and 6h, with GraphSAGE-LSTM achieving the lowest RMSE at the 6-hour horizon at both sites.

4.6. Uncertainty decomposition: aleatoric vs. epistemic

Table 7 will report the aleatoric/epistemic decomposition (Eq. 5) for the best-performing backbone at each site-horizon combination, together with the combined predictive interval’s empirical coverage at nominal levels 80%, 90%, and 95%. σ_{ale} (W/m^2) is the heteroscedastic-head estimate of irreducible cloud-stochasticity noise, averaged over the T SGLD posterior samples; σ_{epi} (W/m^2) is the spread of the T posterior point predictions and shrinks as more representative training data is available; their ratio $\sigma_{\text{epi}}/\sigma_{\text{total}}$ indicates what fraction of total predictive uncertainty is, in principle, reducible by further training.

The backbone for El Paso is GraphSAGE-LSTM (best overall model at 3h and 6h in the fixed-graph baseline, Table 5); for Uniandes the best deep-learning backbone per horizon is used (ResNet-LSTM at 1h, GraphSAGE-LSTM at 3h and 6h). The backbone selection will be revisited once the graph-construction ablation (Table 6) completes, in case the weighted k -NN variant supersedes the fixed-graph baseline used here.

5. Discussion

5.1. Why does GraphSAGE outperform ResNet at longer horizons?

At the 6-hour horizon for El Paso, GraphSAGE-LSTM achieves a 2.2-percentage-point skill improvement over ResNet-LSTM (0.636 vs. 0.614). We hypothesise two mechanisms:

Irregular spatial structure. At longer lead times, the relevant cloud signal migrates from the centre of the patch (where it directly causes GHI reduction) toward the periphery, where incoming cloud advection is reflected in the satellite channels. CNN filters, being translation-equivariant, apply the same spatial template across all positions; GraphSAGE, by contrast, learns separate aggregation rules that can weight distant pixels differently depending on their graph neighbourhood. This flexibility may be especially valuable for detecting cloud-edge gradients that precede GHI events.

El Paso (César) — 6-hour forecast, test set sample (Jan 2024)

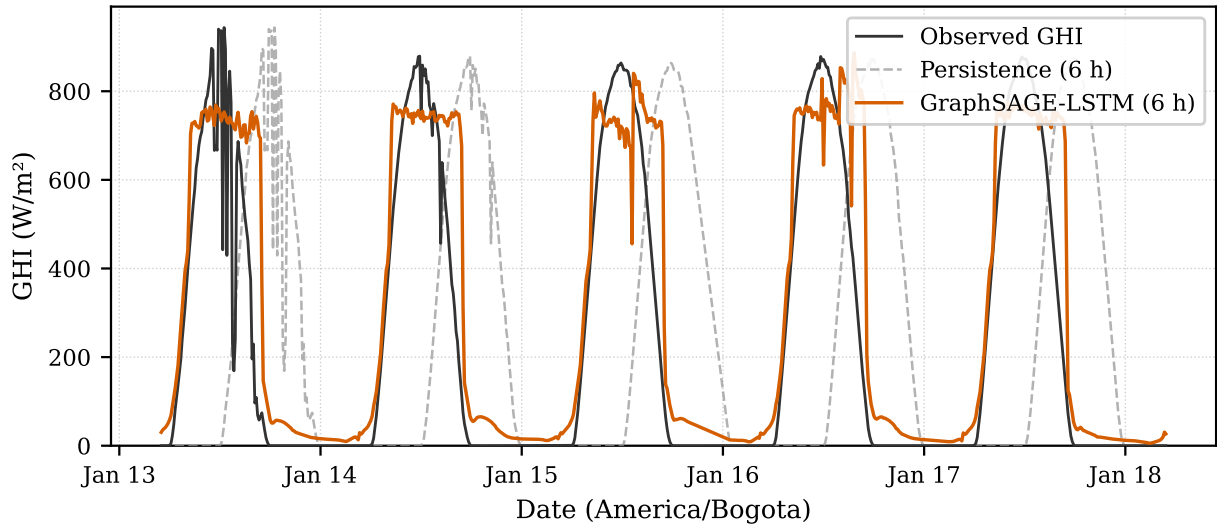


Figure 3: Five-day excerpt of the 6-hour GHI forecast from the best model (GraphSAGE-LSTM Optuna, seed 42) at El Paso in January 2024. Persistence (dashed) simply repeats the GHI observed 6 hours prior, while GraphSAGE-LSTM anticipates the diurnal rise and cloud events. Timestamps in local time (America/Bogotá, UTC−5).

Table 7

Aleatoric/epistemic uncertainty decomposition. σ_{ale} , σ_{epi} , σ_{total} in W/m^2 (Eq. 5); cov_{day} : empirical daytime coverage of the combined interval $\bar{\mu} \pm z_{1-\alpha/2} \sigma_{\text{total}}$ on the test set. Backbone: GraphSAGE-LSTM for El Paso; best-per-horizon deep-learning backbone for Uniandes (ResNet-LSTM at 1h, GraphSAGE-LSTM at 3h and 6h). [SGLD sampling and variance-head training in progress; values to be populated as runs complete.]

Site	H	σ_{ale}	σ_{epi}	σ_{total}	$\sigma_{\text{epi}}/\sigma_{\text{total}}$	cov_{day} (90%)
El Paso	3h	—	—	—	—	—
El Paso	6h	—	—	—	—	—
Uniandes	1h	—	—	—	—	—
Uniandes	3h	—	—	—	—	—
Uniandes	6h	—	—	—	—	—

Reduced redundancy. The ResNet applies multiple stacked convolutions that effectively redundantly process the spatial information across layers; the GraphSAGE message-passing is applied once per hop and is less prone to over-smoothing at the patch scale ($P = 16$). The lower seed variance of GraphSAGE (± 0.013 vs. ± 0.023 at El Paso 6h) is consistent with a more stable loss landscape.

5.2. Why is forecasting skill lower at Uniandes than at El Paso?

Across all deep-learning models and horizons, absolute skill is markedly lower at the Uniandes mountain site than at El Paso (e.g., GraphSAGE-LSTM at 6h: $\text{Skill}_{\text{day}} = 0.417$ vs. 0.636). We attribute this gap to two related properties of the Bogotá climate:

Diurnal regularity narrows the room for improvement. At 4.6°N, the solar declination varies little across the year ($\pm 4.6^\circ$), producing a highly repeatable clear-sky envelope. Cloud formation in Bogotá is predominantly triggered by orographic lifting in the early afternoon, generating a strong, regular diurnal cycle. Much of the predictable structure in the Uniandes GHI series is therefore already encoded in its own recent history and time-of-day, leaving comparatively less *additional* information for a 4-hour satellite patch sequence to contribute — which compresses the achievable skill ceiling for every model, spatially-aware or not.

Satellite signal saturation at longer horizons. At 6 hours, the satellite patch observed at t may carry little predictive information about the ground GHI at $t + 6$ h: the cloud system that will eventually affect the site has often not yet entered the 16×16 -pixel field of view, particularly at a site where convection forms locally through orographic lifting rather than advecting in from a stable upstream direction. This limits how much spatial encoding — regardless of architecture — can contribute at the longest horizon.

This result suggests that *site-specific climatological structure should inform both model design and expectation-setting*: in climates with strong, regular diurnal cloud patterns and locally-forming convection, the marginal benefit of spatial deep learning over simpler temporal signals is smaller, and reported skill scores should always be interpreted relative to the local persistence baseline rather than compared at face value across sites.

5.3. Why is the 1-hour horizon intractable at El Paso?

The negative skill at 1-hour El Paso (all DL models, $\text{Skill}_{\text{day}} \approx -0.17$ to -0.20) can be explained by the *low persistence error*: at 204.4 W/m², persistence is already very accurate because the predominantly clear-sky regime means irradiance rarely drops by more than a small fraction over 10 minutes. The DL models, trained to minimise MSE over the full distribution (including the occasional cloud event), learn a slightly smoothed version of the target trajectory that incurs higher error on the flat clear-sky majority.

A potential remedy is a loss function that differentially weights cloud events relative to the clear-sky envelope (e.g., scaling the per-sample MSE by the departure of observed GHI from the clear-sky reference), or dedicated cloud-event detection as a preprocessing step.

5.4. Practical implications

- For **El Paso-type sites** (high irradiance, moderate variability), GraphSAGE-LSTM with Optuna-tuned hyperparameters is the recommended model at 3h and 6h horizons, with a 60+% reduction in RMSE over persistence. At 1h, the system should fall back to persistence or a light-weight nowcasting model.
- For **Uniandes-type sites** (tropical mountain, high convective variability), even the best models achieve markedly lower skill than at El Paso ($\text{Skill}_{\text{day}} \leq 0.42$ vs. 0.64 at 6h); operators should calibrate reserve margins to the local persistence baseline rather than assume that skill levels transfer across climates. GraphSAGE-LSTM remains the most consistent top performer at 3h and 6h, but gains over persistence are smaller and more sensitive to careful hyperparameter tuning.
- The high seed variance of ResNet-LSTM at El Paso 3h (± 0.090) suggests that multiple retraining runs should be averaged for production deployment, or that GraphSAGE-LSTM should be preferred for its higher stability.

5.5. Disentangling uncertainty: why aleatoric and epistemic matter separately

Grid operators require probabilistic outputs — prediction intervals that are well-calibrated across conditions of varying cloud cover — not just point estimates. Beyond calibration, however, the central premise of this work is that knowing the *source* of forecast uncertainty is itself operationally valuable.

Why a single symmetric interval is not enough. A single, undifferentiated prediction interval — however it is calibrated — reports one half-width per forecast and does not distinguish *why* a given forecast is uncertain. Two forecasts can carry the same nominal interval width for entirely different operational reasons: one because the site is entering a genuinely volatile but well-characterised convective afternoon (aleatoric), the other because the model has rarely seen the current satellite-imagery pattern during training (epistemic). An operator cannot tell these apart from the interval alone, yet the appropriate response differs — accept the irreducible variance in the first case; flag the forecast as low-confidence and prefer a conservative fallback (e.g. persistence, or a wider manually-set margin) in the second.

Aleatoric uncertainty as a proxy for cloud-regime volatility. We hypothesise that the heteroscedastic variance head (Eq. 3) will assign larger σ_{ale} to inputs preceding rapid convective transitions — the regime that, on general

grounds, is hardest to forecast deterministically at both sites — and smaller σ_{ale} during stable clear-sky mornings, providing a continuously-varying, input-dependent signal of intrinsic forecast difficulty that a constant $\hat{q}$ cannot express. This is a design hypothesis, not yet an empirical finding (Section 5, Limitations); Section 4 does not currently attribute forecast errors to specific cloud regimes.

Epistemic uncertainty as a data-quality and deployment-readiness signal. Because σ_{epi} (Eq. 5) reflects disagreement among SGLD posterior samples, it is expected to be larger for satellite-imagery patterns under-represented in the training period (e.g. rare cloud morphologies, seasons with sparse data) and to shrink as the training set grows — without requiring the variance head itself to change. This makes σ_{epi} a natural diagnostic for *when* a forecasting pipeline should be retrained or supplemented with additional data at a given site, complementing the purely architecture-level comparison of Sections 3.3.3–4.3.

5.6. Implications for solar-integrated power system operation

The results presented here connect directly to operational decision-making in solar-integrated power systems.

Situational awareness under cloud-induced uncertainty. GraphSAGE-LSTM, by learning spatially-aware representations of incoming cloud systems from satellite imagery, provides operators with earlier detection of impending irradiance ramps — the primary source of generation-side uncertainty in PV-heavy grids. The 63.6% RMSE reduction over persistence at 6h for El Paso translates to significantly tighter dispatch intervals and reduced reserve requirements.

Disentangled intervals for risk-stratified dispatch. The aleatoric/epistemic decomposition (Section 3.6) converts any point forecast into both an interval and a diagnosis of *why* it is wide. For a system operator, this supports two distinct, complementary actions tied to the same forecast: (i) sizing reserve margins to the combined predictive interval $\bar{\mu} \pm z_{1-\alpha/2} \sigma_{\text{total}}$ for routine risk management; and (ii) flagging forecasts with a high $\sigma_{\text{epi}}/\sigma_{\text{total}}$ ratio as candidates for a more conservative fallback strategy or for prioritised inclusion in future retraining data, a distinction unavailable from a single undifferentiated interval.

Climate- and horizon-adaptive model selection. The marked difference in absolute skill between El Paso (up to $\text{Skill}_{\text{day}} = 0.636$ at 6h) and Uniandes (up to 0.417) has a direct operational implication: forecasting pipelines should be benchmarked and calibrated locally rather than assuming that skill levels transfer across climates. Within a site, the best architecture also depends on the horizon — persistence is hard to beat at 1h in the clear-sky-dominated El Paso regime, while GraphSAGE-LSTM is the most consistent top performer at 3h and 6h at both sites. A horizon-aware portfolio strategy — a light-weight nowcasting fallback at very short horizons, spatial deep learning at multi-hour horizons — could optimise the accuracy-cost tradeoff for operational deployment.

5.7. Limitations

Geographic scope. Both sites are located in Colombia and represent two tropical climate sub-types. Generalisation to other tropical or subtropical regions requires additional validation.

Flat MLP ablation in progress. The FlatMLP experiments are currently completing their Optuna runs (16 of the 24 site $\times$ horizon $\times$ seed configurations evaluated at the time of writing). Final results will be incorporated to quantify what fraction of deep-learning skill arises from spatial encoding versus temporal dynamics alone.

Graph-construction ablation uses fewer seeds. The weighted k -NN GraphSAGE-LSTM variant (Table 6) is evaluated with two random seeds rather than four, for the compute-budget reasons described in Section 3.4. This narrower seed budget should be kept in mind when comparing its mean skill against the four-seed fixed-graph baseline; the comparison should be treated as indicative pending a larger seed budget.

Uncertainty decomposition has not yet been empirically validated. The SGLD sampling and variance-head training pipeline (Section 3.6) is implemented but the resulting aleatoric/epistemic decomposition (Table 7) has not yet been evaluated on the test set. In particular, the claim that σ_{epi} tracks data scarcity and that σ_{ale} tracks cloud-regime volatility (Section 5.5) is currently a hypothesis motivating the design, not yet an empirical finding; both will be assessed quantitatively once results are available.

Single-step output. Current models produce a single-step forecast at horizon H . Simultaneous multi-step output (forecasts for all three horizons from a single forward pass) could improve consistency and reduce inference cost in real-time operational forecasting pipelines.

6. Conclusion

We presented a graph neural network framework for satellite-based solar irradiance forecasting that addresses the situational-awareness needs of solar-integrated power system operators in tropical climates, with a focus on *robust*,

uncertainty-aware forecasting rather than point accuracy alone. The key contributions are: (i) a GraphSAGE-LSTM architecture that explicitly models spatial cloud structure from GOES-16 patches, together with an ablation isolating the value of learned, distance-weighted spatial connectivity over a fixed local graph; (ii) a rigorous comparative evaluation against a ResNet-LSTM baseline, a flat-MLP spatial-ablation baseline, and persistence at two Colombian sites with contrasting climatic regimes; and (iii) a disentangled aleatoric/epistemic uncertainty-quantification layer — combining a heteroscedastic Gaussian variance head with Stochastic Gradient Langevin Dynamics (SGLD) posterior sampling — that tells an operator not just how uncertain a forecast is, but *why*, supporting risk-stratified operational responses that a single undifferentiated interval cannot.

GraphSAGE-LSTM (fixed-graph variant) achieves the best skill among the deep-learning models at the 6-hour horizon at *both* sites (El Paso: $\text{Skill}_{\text{day}} = 0.636$ vs. 0.614 for ResNet-LSTM; Uniandes: 0.417 vs. 0.368), with lower seed variance throughout, confirming that graph-structured neighbourhood aggregation on the pixel graph provides a more stable and expressive spatial representation than fixed convolutional filters even without learned edge weighting. Whether the learned, distance-weighted k -NN graph variant improves further on this result is the subject of an ongoing ablation (Table 6).

At the **tropical mountain site** (Uniandes), however, the absolute skill of every model is markedly lower than at El Paso even at the most favourable horizon ($\text{Skill}_{\text{day}} \leq 0.42$ vs. 0.64 at 6h), revealing that the strong diurnal autocorrelation of the Bogotá clear-sky cycle and the locally-driven nature of orographic convection limit the marginal benefit of satellite spatial features at this site. This finding has a direct operational implication: model selection and skill expectations should be climate-aware, and forecasting pipelines should always be benchmarked and calibrated against the local persistence reference rather than assumed to transfer across sites.

At the **1-hour horizon**, no model consistently beats persistence at El Paso, consistent with the low persistence error (204.4 W/m²) in a predominantly clear-sky regime. Modest positive skill (≈ 0.15) is achieved at Uniandes, where frequent cloud passages create exploitable structure.

Toward robust, disentangled forecasting. The Gaussian negative-log-likelihood variance head and SGLD posterior sampling pipeline (Section 3.6) are implemented; remaining work is to complete sampling across all site-horizon backbones and empirically evaluate the resulting decomposition (Table 7) against the hypotheses motivating its design (Section 5.5): that σ_{ale} tracks cloud-regime volatility and σ_{epi} tracks data scarcity. A natural extension is to use σ_{epi} as an active-learning signal — prioritising under-represented conditions for future data collection — closing the loop between uncertainty quantification and dataset improvement.

Additional planned work includes completing the FlatMLP spatial-ablation sweep and the weighted k -NN graph-construction ablation (Table 6) across the full seed budget, multi-step joint-horizon outputs, and extension to further Latin American sites to build a broader tropical solar forecasting benchmark with disentangled, operationally actionable uncertainty quantification.

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